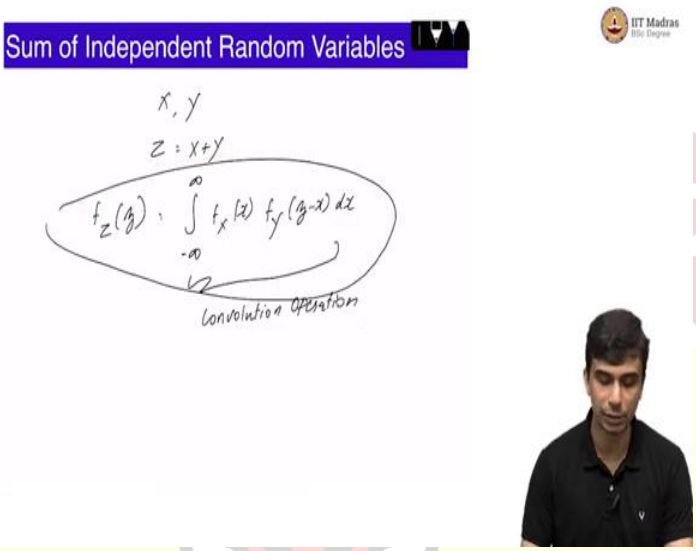


IIT Madras

ONLINE DEGREE

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Independent Random Variables

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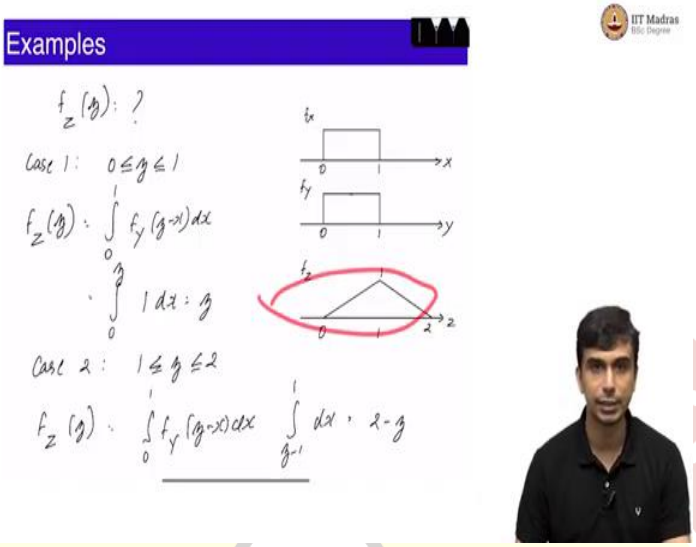


Now, that we have defined independence of random variables, we can see some interesting operations that can be performed on these random variables. These are pretty elementary operations and how in practice several simple random variables are combined to create more complex random variables and one of the most basic ways to combine random variables is addition.

So, that sum of independent random variables you have x and y are independent random variables and let us say Z is equal to X + Y. In the same way that we did for discrete random variables we can also show that the density of this new random variable Z is given by $\int_{-\infty}^{\infty} f_X(x)\, f_Y(z-x)\, dx$, this operation is called the convolution operation.

And is exactly analogous to the discrete random variable case, the only difference is that the summation there is replaced by the integration dx here. I am not going to prove this statement here but the proof is almost the same as you had for the discrete random variable case. However, this seems a pretty complex expression we will see a very for a special case we can work this out and see how this, what this turns out to be.

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Let us take the simplest special case that you can think of where X and Y are uniform in the range of 0 to 1, right. So, X and Y are only form the range of 0 to 1 in that case what will be the density of Z? We can immediately see that the density of Z is going to be 0 outside 0 to 2, why is that? Because you are adding two numbers between 0 to 1 the z is going to be between 0 to 2 right.

So,$f_Z(z)$ is this the question is what is f z of z; that is the question and the outside 0 to 2 is going to be 0 that is obvious. Now, between 0 to do let us make it two cases; this case 1 is z is between 0 to 1, in between 1 to 1 what is $f_Z(z)$? $f_Z(z)$ is $\int_{-\infty}^{\infty} f_X(x)\, f_Y(z-x)dx$ well you do not need to take $(-\infty\ to\ \infty)$ because f (x) is 0 outside 0 to 1.

So, we can immediately change this to 0 to 1 and f x of x within the range of 0 to 1 is simply 1, so you can basically just remove that also. So, you have integral 0 to 1 $f_y(z-x)$ dx and for what values of X will $f_y(z-x)$ being non-zero? $f_y(z-x)$ recall, f y can only take is only taking two values either 1 or 0, it is taking a value of 1 in the range of 0 to 1 and 0 outside.

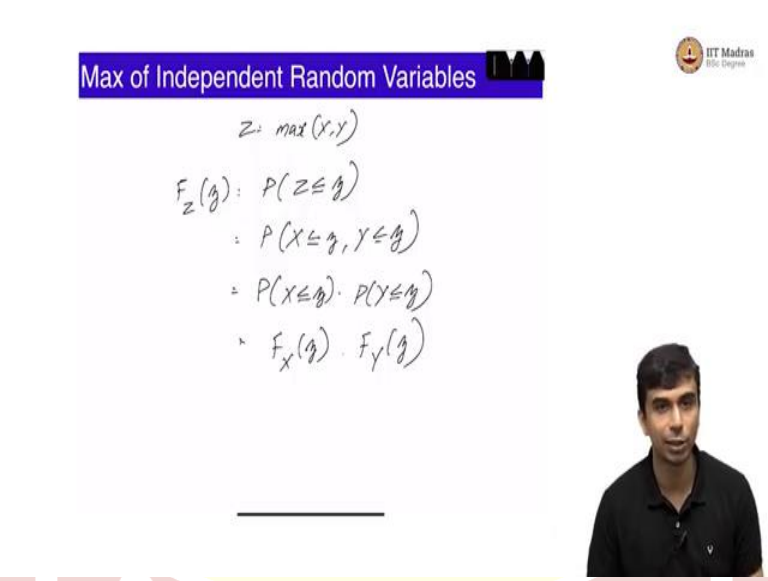
So, F when will $f_y(z-x)$ be non-zero; well $f_y(z-x)$ be non-zero when (z – x) is between 0 to 1, so or in other words x is between 0 to z, integral 0 to z f y of z - x dx but when whenever f y of z - x is non-zero it is exactly equal to 1. So, we just replace that by 1. So, which means that f z of z is equal to z. This is true only for z in the range of 0 to 1.

However, for z in the range of 0 to 2 it is we can work this out in the same way and get the following case 2; which is between 1 to 2. $f_Z(z)$ this integral $\int_0^1 f_Y(z-x)dx$, so when z is in the range of

1 to 2 x, mean when will $f_Y(z-x)dx$ be non-zero? Well $f_Y(z-x)dx$ be non-zero when (z – x) is in the range of 0 to 1 and when z is in the range of 1 to 2 this can happen only when x is in the range of (z - 1) to 1.

Intuitively also it makes sense only when x is in the range of z - 1 to 1 y can I mean when you add with y you can get back z, if x is less than (z – 1) there is no way you can by adding y you can get z. So, that is the logic here, so x is in the range of z - 1 to 1 in that range $f_Y(z-x)$is exactly equal to 1, so this is equal to 2 - z, so putting all of that together we can say that fz of z is 0 in the range of outside 0 to 2 and within 0 to 1 it is equal to z and within 1 to 2 it is 2 - z. So, this results in the density of Z having this nice triangular shape and you can easily check that this also integrates to 1 because the area of this triangle is exactly equal to 1. So, this is a simple example of sum of two independent random variables.

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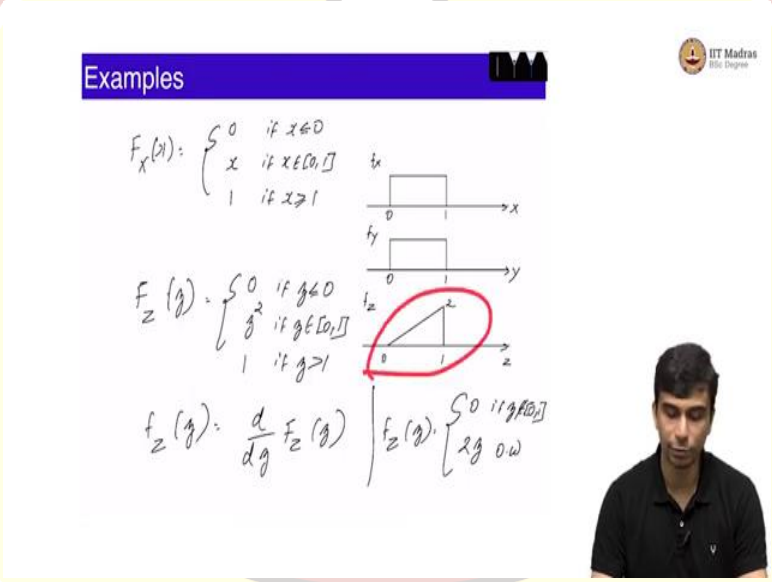
Another operation that is often performed is the max operation. You can have X and Y are let us say two independent random variables and Z is the maximum of X and Y for this particular operation you can use the tool of the cumulative distribution function. What is the cumulative distribution function of Z? $f_Z(z)$ is probability of capital Z less than or equal to Z but Z is equal to max of X, Y.

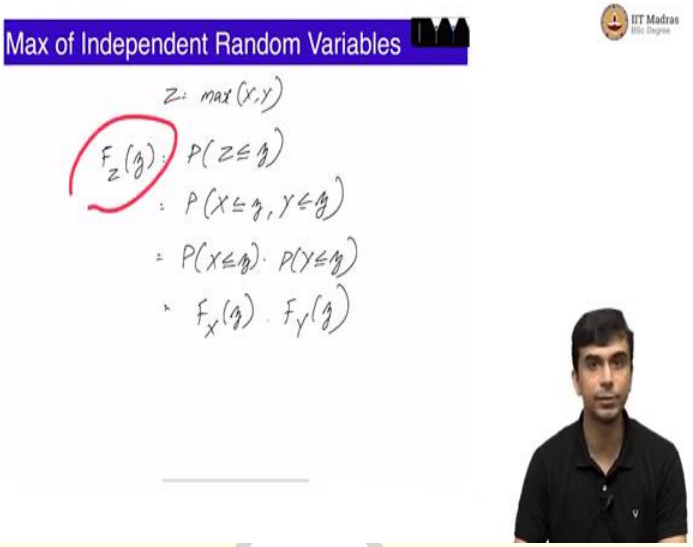
When is the max of two numbers less than or equal to Z? It is if and only if both of these numbers are less than or equal to Z. So, this is equal to probability of X less than or equal to Z sorry and Y

less than or equal to Z. But we have that X and Y are independent random variables so this is exactly equal to probability of X less than equal to Z into probability of Y less than or equal to Z.

Probability of X less than or equal to Z that is exactly equal to $f_X(z)dx$ and probability of Y is an equals it is exactly equal to f Z, $f_Y(z)$, so this is a much nicer situation where you have the cumulative distribution function of Z is simply given by the product of the CDFs, from the CDF how will you get the density? It is pretty simple; you just differentiate the CDF to get the density why is that? Because by integrating the density you get the CDF, by differentiating the CDF you get the density, this is a pretty basic property that comes from the definition of the CDF.

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Let us apply this for a very simple example once again the same example where X and Y are independent uniform 0 1, so if for this particular case you have F_X is simply equal to 0, if X is less than or equal to 0, X, if X belongs to 1 ,, 0 , 1 and 1 if X is greater than or equal to 1. So, then you have it because x and y are the same distribution f X and f Y are the same even though x and y are independent random variables f x and f y are the same.

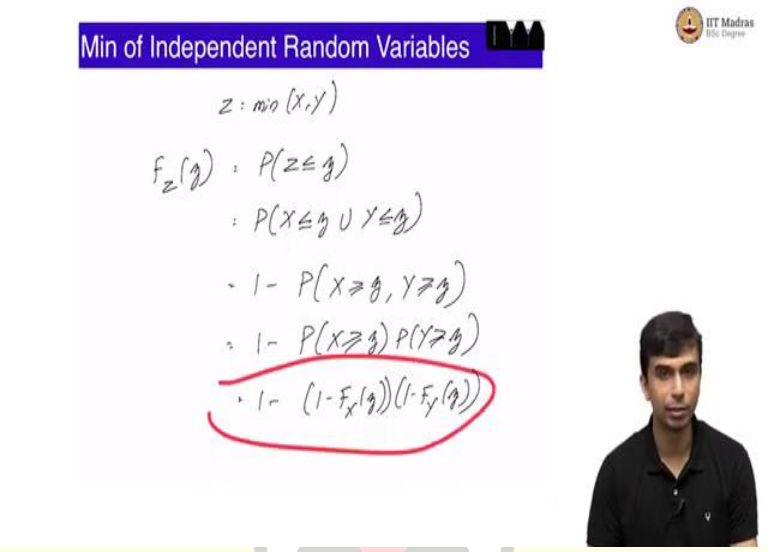
Now, what is F_Z? F_Z is simply F_X (z) × F_Y(z), the CDF of Z is simply F_X(x) × F_Y(y) so based on this we have F_Z(z), f Z of Z is equal to this is F_X(x), F_Z(z) is simply 0 into 0 if Z less than equal to 0, its 0 if z is less than or equal to 0 equal to x z squared if Z is in the range of 0 to 1 and 1 if Z is greater than 1, this is the CDF of z which comes from the simple logic that we derived here.

The CDF of Z is simply the product of the CDFs of x and y. So, that is fine that is have the CDF but how do you get the density function from the CDF; that is you can just get the density function on the CDF by differentiating, so F_Z(z), is simply d by d z of F_Z(z), so this is a pretty basic property its slightly more advanced property than just the simple bounds but it is also obvious to see that this is the case.

So, by differentiating this function we get that F, the density function F_Z(z), is equal to 0 if z does not belong to 1 , 2. Within the range of sorry it is not like 0 , 1 sorry, if Z does not belong to 0 , 1. Within the range of 0 to 1 you differentiate z squared you get 2z so which would result in

your density function looking like this nice triangular function. So, this is two z we can once again check that this also is a valid density function because the area of this triangle is equal to 1.

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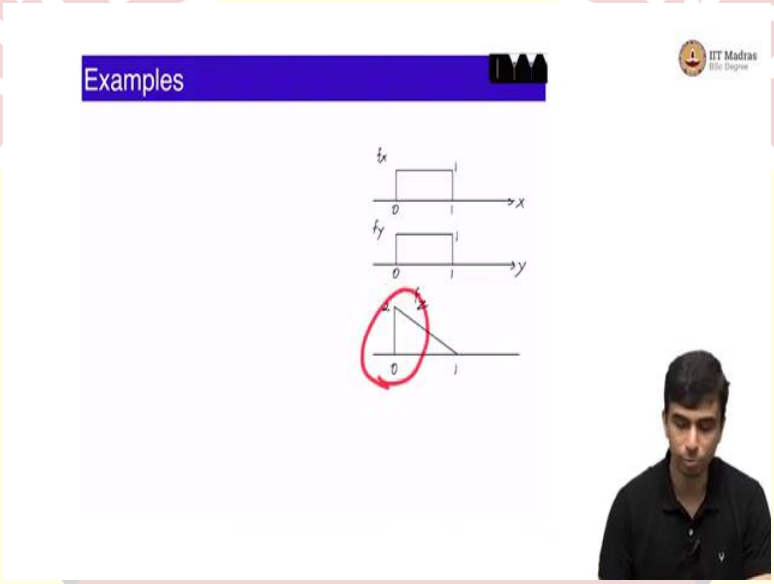
You can do a very similar logic for min I am not going to work out the full details here but min requires a slight bit of a tricky transformation which is you have X and Y are two independent random variables and if Z is the min of X , Y instead of giving the cumulative distribution of z will do this, what is the cumulative distribution of set Z, $F_Z(z)$,? Is probability of Z is less than or equal to Z.

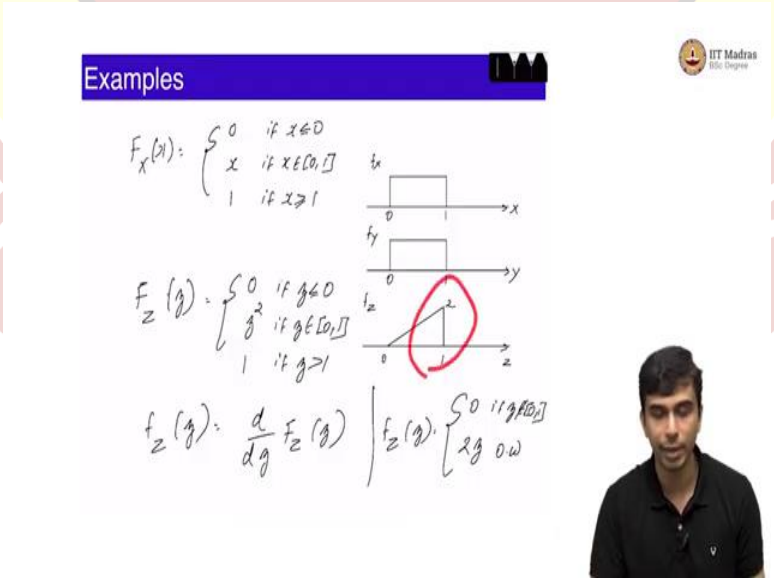
When is the minimum of two variables less than equal to Z; when at least one of them is less or equal to Z so which is probability of X less than or equal to Z Union Y less than or equal to Z, so this is the difference here. In the case of maximum, it was und intersection, in the case of minimum when is the minimum of two random variables less than equal to Z when at least one of them is less than equal to z.

So, the union is always harder to handle with independent random variables than intersection but here is where you can use the De Morgan's Laws; probability of that is A union B is simply A complement intersection B complement the whole complement, so which is we can just use 1 - probability of X greater than or equal to Z and Y greater than or equal to Z.

So, this is a pretty simple application of De Morgan's Laws which is but X and Y are independent so and that means that probability of X greater than equal to Z and Y greater than equal to Z is simply probability of X greater than or equal to Z and probability of into probability of Y greater than or equal to Z. And how can you relate this to the CDF of X and Y? Probability of X greater than equal to Z is simply exactly equal to 1 - probability of x less than or equal to z which is equal to 1, $F_X(z)$. So, it is 1 -, $1 – F_x(z)\times\{ 1 – F_Y(z)\}$,.

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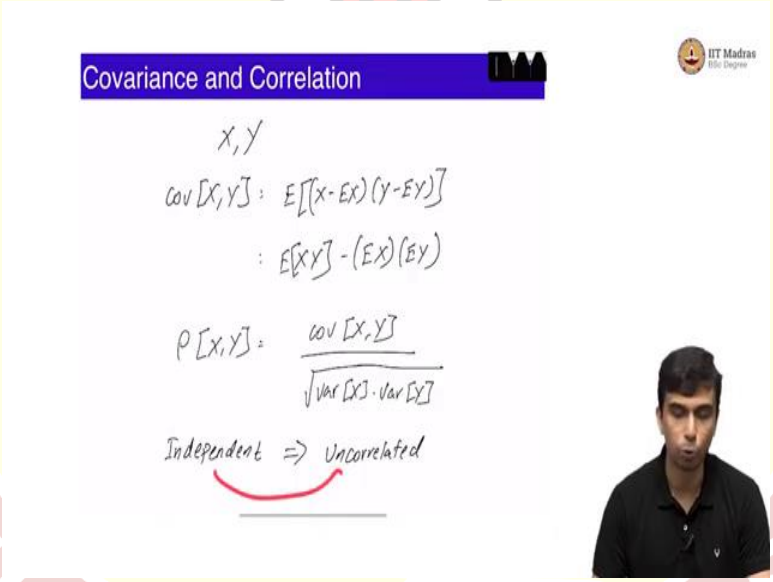


You can simplify this even further but at this point you can stop you can apply this rule for example to get the answer for this example that we have been using throughout which is X and Y are both

uniform 0, 1 random variables and your density of z will look like this instead, there is an excise where you can compute the density of the minimum of two random variables and of these two random variables in particular where x and y are uniform 0 1 and you can get the density of Z.

And intuitively it makes sense that the minimum of two random variables is closer to 0 here and the maximum of two random variables here was the density of that was higher, so you had X and Y by uniform 0 1, so the maximum of that the density of this random variable z is higher when near 1 and lower when we are 0, minimum is exactly the reverse where the density of this minimum of the two random variables is higher near 0 and lower near 1. So, with that we are done with some basic operations, there are more complicated operations but these are the elementary operations on random variables, based on which other operations.

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Now, let us move to the next topic of covariance and correlation. Let us define the covariance of two random variables, let us say you have X and Y. We have already defined this for the case of discrete random variables it is exactly the same definition. Covariance of two random variables X and Y is simply exactly equal to E[(X-EX)(Y-EY)]], this we have already seen is exactly equal to E[XY]- (EX)(EY), this is just a simplification using linearity of expectation. This definition applies also for continuous random variables and it is a very similar object.

You can also define a related object called the correlation coefficient which is often simply written as rho of x ,y and rho of X ,Y is in a sense a normalized version of the covariance

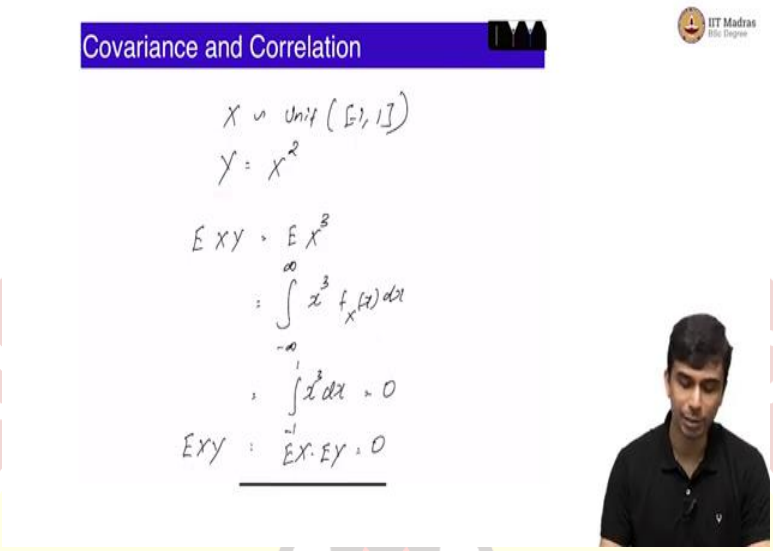
which is $\frac{\text{covariance of [X ,Y]}}{\sqrt{Var[X] \times Var[Y]}}$.

Why is rho more useful? I mean why is rho more intuitive than covariance? That is because rho is bounded between - 1 and + 1, if rho is equal to - 1 then y is exactly equal to - x. When rho equal to + 1 then x is equal to y so when rho equal to 0 means cov [x , y] is 0, so that is the we already seen the definition of uncorrelated random variables.

We will say two random variables are uncorrelated if covariance of [x , y] is 0 and that definition holds here also, this is we can view correlation coefficient as a normalized covariance will not go any deeper than this and one consequence we have seen in the case of discrete random variables is that independence implies uncorrelation right so we already seen that is independence, if we have two random variables which are independent then they are uncorrelated meaning that their covariance is 0.

Independent implies uncorrelated, why is that? Let us follows from the definition of independence expected value of [X Y] is exactly equal to (EX)× (EY), so this covariance is going to be equal to 0. But the other way around is not true, uncorrelated does not imply that independence. Here is a we have already seen an example for the case of discrete random variables, here is an another example for continuous random variables.

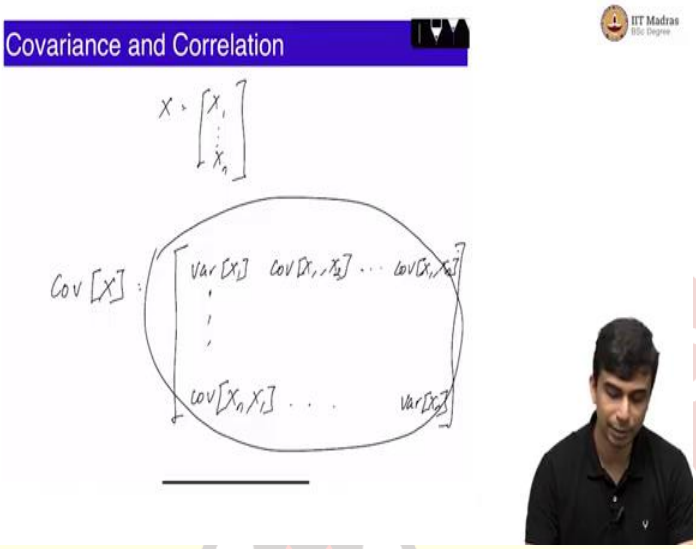
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Let us say you have X is uniform in the range of - 1 to 1 so distributor is uniform. In the range of - 1 to 1, we already seen this density several times. Let us call $Y = X^2$, Y is equal to X square you can easily see that X and Y are not independent. But what is E[xy]? E[xy] is exactly equal to EX^3 and EX^3 we can write as the $\int_{-\infty}^{\infty} x^3 f_X(x)\, dx$ which is equal to $\int_{-\infty}^{\infty} x^3 dx$ and because x^3 is an odd function is equal to 0, so we have E_{XY} is equal to 0 and EX is also equal to 0 or is that x is uniform - 1 to 1, so Ex is, so that we have that E_{XY} is equal to(EX) ×(EY) which is equal to 0.

So, we have that, in this case X and Y are uncorrelated random variables but it is clearly obvious that X and Y are dependent random variables if $y = x^2$ you cannot say it is obviously true that x and y are dependent random variables. So, here is a simple example of the case where you have two random variables which are uncorrelated but they are dependent. So, you can also have covariance for random vectors, let me just define random vectors first.

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So, you could have instead of just two random variables X and Y you could have a random vector which is let us call that random vector X as with n dimensions x1, x2, x3 so on. You have n random variables we put that together into a vector, we will call that a random vector, any such random vector, any pair of random variables have covariance. So, now we can have what is called a covariance matrix.

Now, you have an N cross N matrix where the first element is covariance of X1, X1, we have already seen that covariance of X, X is exactly equal to the variance of X, so the first element is variance of X1. The second element is covariance of X1 , X2 and so on you have covariance of X1 , Xn this is the nth element and you can go all the way, this would be covariance of Xn , X1.

We have covariance of Xn , Xn which is variance of Xn. So, there is an n×n matrix where each entry the i , jth entry is the covariance of x i , x j, so it is called the covariance matrix of the random vector X. And by definition of covariance it is independent, it is symmetric. So, by the differential covariance its symmetry which means the covariance of X , Y is equal to covariance of [Y , X].

So, this matrix happens to be a symmetric matrix that is, this covariance matrix of X is equal to the its transpose, so this is a covariance matrix of random vector is a fundamental object that we will see studied often and we will actually see examples of it later in this course itself. So, with that we can wrap up expectation and covariance.